

foreign/relative prices in CPI equation

core CPI
(rel. change)

$$\pi = \dots \dots \dots [\pi^* + s - \Delta \bar{z}]$$

↑ to LHS

direct effect $-\pi$ correct for rer trend

$$= [\pi^* + s - (\bar{\pi}^* + \bar{s} - \bar{\pi})]$$

indirect effect (rmc) $[\bar{z} - \bar{z}]$

$= \bar{z} - \bar{z} - \text{gap}$

food CPI apply same logic, with "food rer" = $\pi^{*F} + s - \frac{\pi^F}{\Delta z^F}$

$$\pi^F = \dots \dots \dots [\pi^{*F} + s - \Delta \bar{z}^F]$$

direct effect correct for rer trend

$$= \pi^{*F} + s - [\bar{\pi}^{*F} + \bar{s} - \bar{\pi}^F]$$

$$+ \bar{\pi}^* - \bar{\pi}^*$$

$$+ \bar{\pi}^{core} - \bar{\pi}^{core}$$

$$+ \bar{\pi} - \bar{\pi}$$

$$= \pi^{*F} + s - (\bar{\pi}^{*F} - \bar{\pi}^*) = \Delta \bar{r}p^{*F}$$

$$- (\bar{\pi}^* + \bar{s} - \bar{\pi}^{core}) = \Delta \bar{z}$$

$$+ (\bar{\pi}^F - \bar{\pi}) = \Delta \bar{r}p^F$$

$$- (\bar{\pi}^{core} - \bar{\pi}) = \Delta \bar{r}p^{core}$$

indirect effect

$$\left[\begin{array}{c} l - 2^F - l - 2^{\bar{F}} \\ = l - 2 - gap \end{array} \right]$$

(cp above $-\Delta \bar{2}^F$!)
opposite signs

So

$$\begin{aligned} &+ l - rp^{*F} - gap \quad (= l - rp^{*F} - l - rp^{*\bar{F}}) \\ &+ l - 2 - gap \quad (= l - 2 - l - \bar{2}) \\ &- l - rp^F - gap \quad (= l - rp^F - l - \bar{r}\bar{p}^F) \\ &+ l - rp^{cne} - gap \quad (= l - rp^{cne} - l - \bar{r}\bar{p}^{cne}) \end{aligned}$$